Adama Science and Technology University

Department of Applied Mathematics

Numerical Analysis (Math3201) Worksheet 1

1. Determine the sources of error that arise in computing the surface area of the earth using the formula S=$4\pi r^2$, where r is the radius of the earth.

2. Compute the absolute and relative error in approximating a number x by x^*

a) $x = \pi$ and $x^* = \frac{22}{7}$ b) $x = \sqrt{2}$ and $x^* = 1.414$ c) $x = e$ and $x^* = 2.718$

3. The derivative of $f(x)$ at a particular value of x can be approximately be calculated by

$$f'(x) = \frac{f(x+h)-f(x)}{h} \text{ for } f(x) = 7e^{0.5x} \text{ and } h = 0.3, x = 2. \text{ Find}$$

a) The exact value of $f'(x)$ b) The approximate value of $f'(x)$ c) The absolute and relative errors of this approximation

4. If a number x is approximated by x^*=123.456 and the relative error is 0.1, then what is the possible range of the values x?

5. If $u = \frac{x^2 y}{z}$ and the error in x, y and z are 0.001,0.002 and 0.003 respectively, compute the maximum relative error in u when $x = y = z = 3$.

6. Identify the IEEE standard floating-point numbers corresponding to the following bit strings.

	0000000	00000000000000000000

	0000000	00000000000000000000
	1111111	00000000000000000000

	1111111	00000000000000000000

	0000001	00000000000000000000

7. Compute the largest and smallest positive numbers that can be represented in the 32-bit normalized form.

8. Compute the largest and smallest negative numbers that can be represented in the 32-bit normalized form.

9. Repeat (7) for the 32-bit denormalized form.

10. Repeat (8) for the 32-bit denormalized form.

11. Obtain an interval which contains a positive root of the equation

a) $f(x) = \frac{1}{2}x^3 - 3x^2 + 2x + 1 = 0$ b) $f(x) = \cos x - xe^x = 0$

12. Use bisection method to find a) $\sqrt[3]{19}$ b) $\sqrt[3]{23}$ c) $\sqrt[5]{31}$ in four decimal places with error of tolerance $\in = 0.01$.

13. Find the minimum number of iteration required for the Bisection Method to obtain the root of an equation $f(x) = 0$ on [4,5] with a tolerance of $\varepsilon = 10^{-3}$.

14. Use the Bisection method to find solutions accurate to within 10^{-2} for the following problems:

(a) $x - 2^{-x} = 0$ for $0 \le x \le 1$

(b) $x = \tan x$ for $4 \le x \le 4.5$ (c) $2 + \cos(e^x - 2) - e^x = 0$ for $0.5 \le x \le 1.5$

15. Using Bisection Method solve the following equations:

a) $f(x) = \sqrt{x} - cosx$ on [0,1]. Perform only the first three iterations.

b) $x = \sqrt{5}$ correct to two decimal places.

16. Let $f(x) = -x^3 - cosx$. Taking $x_0 = -1$ and $x_1 = 0$. find x_3 using each of the following methods.

a) False position method b) Secant method

17. Obtain the root of the equation $x^x = 100$ correct to three significant digits, using the Secant method .Given that the root lies between 3 and 4.

18. Let $x^3 + 2x^2 - 2x - 4 = 0$ be a given equation. Then find

a) An iteration function $\emptyset$ so that the iteration $x_{n+1} = \emptyset(x_n)$ converges to the desired root.

b) The root of the given equation correct to three decimal places(use $x_0 = 1$)

19. Perform four iterations of the Newton-Raphson method to obtain the approximate value of $\sqrt[3]{19}$ starting with the initial approximation $x_0 = 2.2$

20. Using Newton-Raphson method solve the following equations

a) $3xe^x = 0$ with absolute error tolerance $\varepsilon = 10^{-5}$. Take $x_0 = 1.5$

b) $x^4 - x - 10 = 0$ correct to three decimal places.

21. Find a bound for the number of iterations needed to achieve an approximation with an absolute accuracy 10^{-4} to the solution of $x^3 - x - 1 = 0$ lying in the interval [1,2].Find an approximation to the root with this degree of accuracy.

22. Given the following equation a) $x^4 - x - 10 = 0$ b) $x - e^x = 0$. Determine the initial approximation to find the smallest positive root. Use these to find to the roots correct to three decimal places with the following methods:

i) Regula-Falsi method,

ii) secant method

iii) Newton -Raphson method,.

iv) Fixed point iteration method